

1. Problem 1

a.

$$E = \frac{n^2 h^2}{8mL^2}$$

$$E = \frac{1^2 \cdot (6.626 \cdot 10^{-34})^2}{8 \cdot 9.109 \cdot 10^{-31} \cdot (10 \cdot 10^{-10})}$$

$$\boxed{E = 6.02 \cdot 10^{-20} \text{ J}}$$

$$E = \frac{6.02 \cdot 10^{-20} \text{ J}}{1.602 \cdot 10^{-19} \text{ J/eV}}$$

$$\boxed{E = 0.376 \text{ eV}}$$

$$E = h\nu$$

$$\tilde{\nu} = \frac{\nu}{c}$$

$$\tilde{\nu} = \frac{E}{hc}$$

$$\tilde{\nu} = \frac{6.02 \cdot 10^{-20}}{6.626 \cdot 10^{-34} \cdot 3 \cdot 10^{-10}}$$

$$\boxed{\tilde{\nu} = 3031 \text{ cm}^{-1}}$$

b.

$$\psi = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

$$\psi^* \psi dL = \frac{2}{L} \sin^2\left(\frac{n\pi x}{L}\right) dL$$

n=2

$$\psi^* \psi dL = \frac{2}{10 \cdot 10^{-10}} \sin^2\left(\frac{2\pi \cdot 5 \cdot 10^{-10}}{10 \cdot 10^{-10}}\right) dL$$

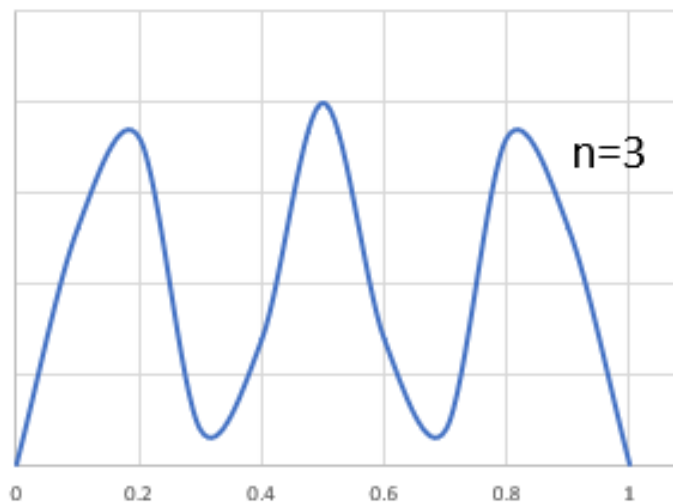
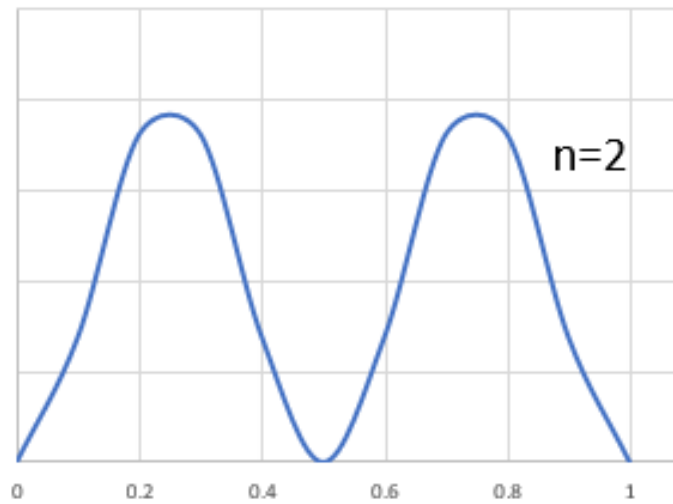
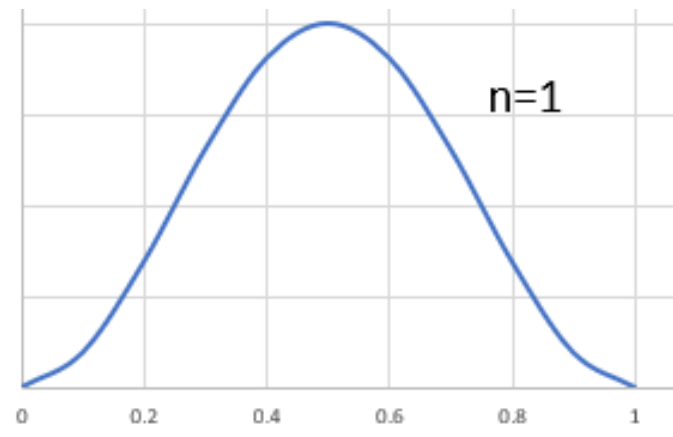
$$\boxed{\psi^* \psi dL = 0}$$

n=3

$$\psi^* \psi dL = \frac{2}{10 \cdot 10^{-10}} \sin^2\left(\frac{3\pi \cdot 5 \cdot 10^{-10}}{10 \cdot 10^{-10}}\right) dL$$

$$\boxed{\psi^* \psi dL = 2 \cdot 10^9 dL}$$

c.



Plots above are $\psi^*\psi$ v Fraction of L. For $n=2$, the probability of an electron being at $0.5L$ is zero. So the probability of the transition from $n=2$ to $n=3$ would be 0.

2. Problem 2 (4.3)

a.

$$\Delta E = E_2 - E_1$$

$$E = \frac{h^2 n^2}{8mL^2}$$

$$\Delta E = \frac{h^2 n_2^2}{8mL^2} - \frac{h^2 n_1^2}{8mL^2} = \frac{h^2}{8mL^2} (n_2^2 - n_1^2)$$

$$E = h\nu$$

$$c = \lambda\nu$$

$$\nu = \frac{c}{\lambda}$$

$$E = \frac{hc}{\lambda}$$

$$\frac{hc}{\lambda} = \frac{h^2}{8mL^2} (n_2^2 - n_1^2)$$

$$\lambda = \frac{8mL^2 c}{h(n_2^2 - n_1^2)}$$

b.

$$\lambda = \frac{8 \cdot 9.109 \cdot 10^{-31} \cdot (10 \cdot 10^{-10}) \cdot 3 \cdot 10^8}{6.626 \cdot 10^{-34} \cdot (2^2 - 1^2)}$$

$$\lambda = 1.10 \cdot 10^{-6} \text{ m}$$

$$\lambda = 1099.84 \text{ nm}$$

3. Problem 3

a. Reduced mass for HBr

$$\begin{aligned}
 m_{\text{H}} &= 1.007875 \text{ amu} \\
 m_{\text{Br}} &= 80.916290 \text{ amu} \\
 \mu &= \frac{m_1 m_2}{m_1 + m_2} \\
 \mu &= \frac{1.007875 \text{ amu} \cdot 80.916290 \text{ amu}}{1.007875 \text{ amu} + 80.916290 \text{ amu}} \\
 \mu &= 0.995427 \text{ amu}
 \end{aligned}$$

b. Find force constant k

$$\begin{aligned}
 \text{Vibration frequency} &= 2558.5 \text{ cm}^{-1} \\
 \omega &= \sqrt{\frac{k}{\mu}} \\
 k &= \omega \mu \\
 \nu &= \frac{c}{\lambda} \\
 \nu &= c \tilde{\nu} \\
 \nu &= 3 \cdot 10^{10} \text{ cm/s} \cdot 2558.5 \text{ cm}^{-1} \\
 \nu &= 7.6755 \cdot 10^{13} \\
 \omega &= 2\pi \nu \\
 \omega &= 2\pi \cdot 7.6755 \cdot 10^{13} \\
 \omega &= 4.823 \cdot 10^{14} \\
 k &= \omega \mu \\
 k &= (4.823 \cdot 10^{14})^2 \cdot \frac{0.995427}{6.022 \cdot 10^{26}} \\
 k &= 384.5 \text{ kg/s}^2
 \end{aligned}$$

c. Given R = 141.4 pm, find moment of inertia

$$\begin{aligned}
 I &= \mu R^2 \\
 I &= \frac{0.995427}{6.022 \cdot 10^{26}} (141.4 \cdot 10^{12})^2 \\
 I &= 3.305 \cdot 10^{-47} \text{ kg}\cdot\text{m}^2
 \end{aligned}$$

d.

$$\hbar = \frac{h}{2\pi} = 1.0547 \cdot 10^{-34} \text{ J}\cdot\text{s}$$

$$B = \frac{\hbar^2}{2I}$$

$$B = \frac{(1.0547 \cdot 10^{-34} \text{ J}\cdot\text{s})^2}{2 \cdot 3.305 \cdot 10^{-47} \text{ kg}\cdot\text{m}^2}$$

$$\boxed{B = 1.683 \cdot 10^{-22} \text{ J}}$$

$$\nu = \frac{E}{h}$$

$$\nu = \frac{1.683 \cdot 10^{-22} \text{ J}}{6.626 \cdot 10^{-34} \text{ J}\cdot\text{s}}$$

$$\boxed{\nu = 2.54 \cdot 10^{11} \text{ s}^{-1}}$$

0 to 1 is at 2B

$$\nu = 2 \cdot 2.54 \cdot 10^{11} \text{ s}^{-1}$$

$$\boxed{\nu = 5.08 \cdot 10^{11} \text{ s}^{-1}}$$

1 to 2 is at 4B

$$\nu = 4 \cdot 2.54 \cdot 10^{11} \text{ s}^{-1}$$

$$\boxed{\nu = 1.016 \cdot 10^{12} \text{ s}^{-1}}$$

4. Problem 4 (4.9)

a.

$$\omega = \sqrt{\frac{k}{\mu_m}} = 2\pi\nu$$

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu_m}}$$

$$\nu = \frac{1}{2\pi} \sqrt{\frac{1903.17}{1.1385 \cdot 10^{-36}}}$$

$$\boxed{\nu = 6.5072 \cdot 10^{13} \text{ s}^{-1}}$$

$$\tilde{\nu}_{\text{HO}} = \frac{6.5072 \cdot 10^{13} \text{ s}^{-1}}{3 \cdot 10^{10} \text{ cm/s}}$$

$$\boxed{\tilde{\nu}_{\text{HO}} = 2170.56 \text{ cm}^{-1}}$$

b. The value we get for $\nu_{\text{HO}} = 2170.56 \text{ cm}^{-1}$ while the spectroscopically measured one is $\nu_{\text{HO}} = 2143.36 \text{ cm}^{-1}$. Our value is larger than the measured, and the reason for this difference is due to the anharmonicity of CO. The spaces between $n=0$ and 1 are closer than expected for harmonic oscillators.

5. Problem 5 (4.12)

a.

$$\Delta E = h\nu$$

$$\omega = \sqrt{\frac{k}{m}} = 2\pi\nu$$

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$\nu = \frac{\Delta E}{h}$$

$$\nu = \frac{4.25 \cdot 10^{-21}}{6.626 \cdot 10^{-34}}$$

$$\nu = 6.41 \cdot 10^{12} \text{ s}^{-1}$$

$$\sqrt{\frac{k}{m}} = 2\pi \cdot 6.41 \cdot 10^{12} \text{ s}^{-1}$$

$$\sqrt{\frac{k}{m}} = 4.03 \cdot 10^{12} \text{ rad/s}$$

$$k = (4.03 \cdot 10^{12})^2 \cdot 3.15 \cdot 10^{-25}$$

$$\boxed{k = 512 \text{ N/m}}$$

b.

$$2\pi\nu = \sqrt{\frac{k}{m}}$$

$$k = (2\pi\nu)^2 m$$

$$k = (2\pi \cdot 0.805)^2 \cdot 20$$

$$\boxed{k = 512 \text{ N/m}}$$

c. The reason it is said that the chemical bond is compared to a mechanical spring, this is in characteristic that the chemical bond acts like a very strong spring. In essence, the spring in (b) holds 20 kg mass.

6. Problem 6 (4.19)

$$\mu_{\text{H}_2} = \frac{1 \cdot 1}{1 + 1} = \frac{1}{2}$$

$$\mu_{\text{HD}} = \frac{1 \cdot 2}{1 + 2} = \frac{2}{3}$$

$$\mu_{\text{O}_2} = \frac{2 \cdot 2}{2 + 2} = 1$$

$$\mu_{\text{HD}} = \frac{4}{3}\mu_{\text{H}_2}$$

$$\mu_{\text{D}_2} = 2\mu_{\text{H}_2}$$

$$\omega = \sqrt{\frac{k}{\mu_m}} 2\pi\nu = 2\pi c\tilde{\nu}$$

$$\tilde{\nu}_{\text{H}} = \sqrt{\frac{3}{4}}\tilde{\nu}_{\text{H}_2}$$

$$\tilde{\nu}_{\text{HD}} = \sqrt{\frac{3}{4}} \cdot 4400$$

$$\boxed{\tilde{\nu}_{\text{HD}} = 3811 \text{ cm}^{-1}}$$

$$\tilde{\nu}_{\text{D}_2} = \sqrt{\frac{1}{2}}\tilde{\nu}_{\text{H}_2}$$

$$\tilde{\nu}_{\text{D}_2} = \sqrt{\frac{1}{2}} \cdot 4400$$

$$\boxed{\tilde{\nu}_{\text{D}_2} = 3111 \text{ cm}^{-1}}$$